

During my lifetime, I believe that medicine will be revolutionized as we gain precise control over molecular behavior. This potential for creativity and interdisciplinary collaboration is what draws me to chemical biology – a field that bridges molecular design with living systems. Through graduate studies at UCSD, I wish to take the first step toward my long-term goal of becoming a professor and leading innovative research.

With a dream to design miraculous drugs that cure diseases, I first joined a lab at China Pharmaceutical University to experience the whole pre-clinical drug development process, including design, synthesis and evaluation. These initial projects, each spanning a few months, built my foundational skills through hands-on experiments and sparked my initial interest in medicinal chemistry. During my sophomore year, I joined Prof. Jinsong Han's lab, where I independently designed and synthesized novel Wee1 kinase inhibitors for improved anticancer efficacy. I analyzed the structure-activity relationships of Wee1 inhibitors, modified the structures, and optimized synthesis routes, honing my organic chemistry techniques and problem-solving abilities.

My interest evolved toward fluorescent sensor when I observed my mentor synthesizing dyes in the same lab. The vibrant colors and artistic feeling captivated me, leading me to a second project in my junior year. Aiming to develop a fluorescent sensor array for **bacterial recognition**, I focused on poly(*para*-aryleneethynylene) (PPE) derivatives with side-chain modification to promote non-specific electrostatic and hydrophobic interactions with diverse bacteria, generating differential fluorescence quenching for pathogen discrimination. As the lead researcher, I designed and synthesized a series of PPEs with alkyl oxides or aromatic structures, tested the sensor array for bacteria detection, and utilized machine learning algorithms for detection in urine samples. My contributions helped our team win a gold award in a Biotech Innovation Competition. **However, I had some burning questions regarding my research interests going forward. Although I was engaged by the technical aspects of solving open-ended problems, I wanted to find something that would really excite me – what is the thing that would get me out of bed every morning? And how could I find it?**

With that overreaching goal in mind, I led a project to develop **a sensor for protease biomarker quantification and point-of-care diagnosis** during my master's program at Fudan University with Prof. Gang Xing and Prof. Lu Wang. The existing approaches to quantify protease in biological samples are limited by suboptimal sensitivity, poor compatibility with **biological samples**, long test times, and high costs. Inspired by the potential of bioluminescence for point-of-care test, I aimed to develop a chemigenetic bioluminescence resonance energy transfer (BRET)-based sensor to address these issues. Without prior advanced biology experience, this marked my first transition from synthetic chemistry to chemical biology. **These sensors sandwich protease cleavage sites between NanoLuc luciferase protein and the tetracysteine tag, using BRET to shift light emission upon proteolysis. In addition, through tandem arrangement of tetracysteine tags, the sensor achieved picomolar sensitivity.** Encouraged by the promising result, I applied it to quantify matrix metalloproteinase-9 (MMP-9) in human plasma, achieving ~90% accuracy in distinguishing colorectal cancer patients from healthy controls. **The most innovative part is that** . This work encountered several challenges when optimizing the sensor protein. I first tested my hypothesis by connecting multiple tetracysteine tags with commonly used GGS linker, but initial results showed no improvement in BRET efficiency. Moreover, relevant papers using this strategy for tandem fluorescent proteins didn't yield promising outcomes. **Realizing that the distance** between two fluorophore might be the key, I designed three different linkers with varying rigidity to tune the distance and orientation. Finally, I discovered that polyproline linkers performed the best, with further optimization of linker length pushing BRET

efficiency to an unprecedented ~90%. My work has culminated in a poster presentation in an international conference, and a first-author manuscript submitted to *Nature Sensors*. Seeing a connection between my work and the real-world impact gives me a rush of excitement that I am contributing to help those patients in need. Through this project, I found myself enjoying both scoping and solving open-ended problems and hope to further improve with additional formal training in graduate studies.

This year, I was able to further extend my interests in protein engineering with Prof. Neal Devaraj at UC San Diego. His group pioneered chemoenzymatic RNA labeling using tRNA guanine transglycosylase (TGT) and recently demonstrated its potential for DNA labeling. However, *E. coli* TGT bears low efficiency and slow kinetics for DNA labeling. This limitation raised my question: could TGT orthologs from other species offer greater substrate promiscuity? After quickly mastering the enzymatic labeling assay in a week, I independently initiated a project to explore TGTs from other species. Through sequence analysis and AlphaFold structure predictions, I selected two promising candidates – *Staphylococcus aureus* and *Chlamydia trachomatis*. I cloned TGT genes into plasmids, expressed and purified new TGTs, used them for in vitro labeling tests, and compared them with commonly used *E. coli* TGT. Ultimately, I discovered *S. aureus* TGT with superior DNA labeling efficiency. My result on these intriguing behaviors of both TGTs have since sparked ongoing structural studies in the lab to map active-site differences. My contribution led to a first-author manuscript submitted to *ChemBioChem*, further building my experience in enzyme engineering and preparing me for independent PhD research in chemical biology.

I believe my experience spanning medicinal chemistry, fluorescence probe, biosensor development, and protein engineering has offered me a unique perspective into practical problems in probing and manipulating biological systems at the molecular level and makes me uniquely qualified to tackle the most daunting questions in this area. I seek to continue pursuing my curiosities at UCSD, where I plan to build a broad, interdisciplinary skillset for developing chemical biology technologies for understanding molecular and cellular network relevant to human disease.

UCSD's leadership in chemical biology, exemplary faculty, and interdisciplinary environment make it an ideal fit for my goals. In particular, **Prof. Neal Devaraj's** research on developing tools for RNA detection and labeling through enzymatic and non-enzymatic methods resonates with my interest in protein engineering and fluorescent sensor development. Building on my hands-on experience in his lab as a visiting student, I am eager to further expand these techniques to advance RNA imaging and manipulation. An opportunity to work with him is extremely appealing to me as a budding chemical biologist.

I believe I bring with me diverse research experience, interdisciplinary skills, and above all, an insatiable desire to learn and excel. I look forward to the next milestone in my life – a PhD in Biochemistry and Molecular Biophysics from UCSD!